Felix D. Gonzalez

Electrical & Computer Engineering · Software, AI Systems & Embedded Controls

Chicago, IL · US Citizen · felix34003@gmail.com
github.com/felix34003 · linkedin.com/in/felix-gonzalezz/

EDUCATION

Illinois Institute of Technology - Dual Degree Program · Chicago, IL

M.Eng., Computer Engineering · Expected 2028 · GPA: 4.0

B.S., Electrical Engineering · Expected 2027 · GPA: 3.4

Relevant coursework: Digital Systems; Circuit Analysis; Power Engineering; Signals and Systems; Computer Architecture and Assembly; Machine Learning; Embedded Systems; VLSI Design.

PROFESSIONAL EXPERIENCE

Honda Motor Company - Software & Data Engineering Intern

East Liberty, OH · May-August 2026

- Built VINVideo, an internal application connecting vehicle identifiers with vehicle history and process-camera footage using Ignition, scripting, SQL and camera-system integration; iterated with quality-team users.
- Built Excel mapping and maintenance-system transformation workflows to reduce manual predictive-maintenance reporting, then handed the pipeline to the team for final integration.
- Analyzed machine telemetry using Python, Jupyter, Pandas and NumPy; developed statistical thresholds and Ignition scripts connected to Microsoft Teams alerts.
- Used AI-assisted development tools for implementation and debugging alongside stakeholder feedback and iterative review.

Syska Hennessy Group - Software & Information Communications Technology Intern

Chicago, IL · June-August 2025

- Developed an automated engineering drawing-review tool using Python, OpenCV, OCR and a web-based results interface to compare project metadata with reference records.
- Designed a deployment approach for colleagues working without administrator privileges and used AI-assisted code generation and debugging during development.
- Contributed to telecommunications, security and audiovisual design in Revit and AutoCAD, combining drawing coordination with practical site observations.

TECHNICAL SKILLS

Software & AI Systems: Python, C++, SQL, PyTorch, OpenCV, OCR, computer vision experimentation, Pandas, NumPy, Jupyter, HTML/CSS/JavaScript, AI-assisted development.

Embedded & Controls: ESP32, Arduino, Raspberry Pi, embedded Linux, ROS 2, PWM, UART, I2C, DroneCAN, sensor/actuator integration, soldering, wiring, oscilloscope and multimeter measurements.

Engineering Tools: Ignition SCADA, Excel, IBM Maximo integration workflows, Git/GitHub, VHDL, Vivado, Revit, AutoCAD.

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ENGINEERING & PROJECT EXPERIENCE

Illinois Tech Rocketry - Avionics Controls Team Lead

Chicago, IL · August 2025-Present

- Coordinated avionics integration across electrical, aerospace and mechanical workstreams, including development milestones and team interfaces.
- Built and troubleshot sensing and actuator circuitry through soldering, wiring and connector integration; developed power distribution for 7.4 V servo loads and 5 V digital electronics.
- Programmed ESP32 airbrake-servo control in C++ using PWM; used oscilloscopes, signal generators and multimeters to inspect control signals, power rails and hardware interfaces.

Astra - Aircraft Power & Avionics Integration

Team aerospace project · 2026

- Contributed power-system design and avionics integration for a team circular-wing aircraft prototype, with work spanning battery monitoring, distribution and peripheral interfaces.
- Developed electrical-system documentation and investigated UART, PWM, I2C and DroneCAN integration; team demo footage shows an airborne MK1 prototype.

Efficient Computer Vision Model Optimization - Illinois Tech

April-May 2026

- Built a modular PyTorch experimentation workflow for MobileNet classification and object detection, supporting configuration, GPU training, recorded comparisons and ablation studies.
- Evaluated classification architecture tradeoffs: a compact MobileNet model used 42.8% fewer parameters with final logged accuracy of 66.21%, compared with 66.49% for the control; a separate full-width attention model reached 68.15% best logged accuracy.
- Investigated anchor clustering, target assignment, detection heads and DIoU-NMS for a MobileNetV2-backed Tiny YOLOv3 detector; recorded validation mAP of 0.595 versus a recreated baseline of 0.536, with approximately 0.9% fewer parameters.

Indoor Mobile Robot - Illinois Tech ECE 441

Perception Lead & Navigation Support · January-May 2026

- Led perception/semantic-understanding work and supported navigation integration within a multidisciplinary robot team.
- Developed ROS 2/YOLO camera-perception interfaces with configurable image intake, structured detections and operator overlays; supported integration with sensing, drive control and wireless teleoperation.
- Worked within onboard compute and networking constraints, separating perception integration from the team's broader navigation and embedded-control responsibilities.

VHDL Digital Circuit Design - Illinois Tech ECE 218

Coursework project family

- Designed VHDL arithmetic, display-decoding and finite-state-machine exercises using Vivado and an Artix-7 FPGA workflow.
- Practiced testbench simulation, synthesis, implementation and hardware programming; debugged logic and physical input/output integration.